

PERRY BEAMER
CURRICULUM VITAE
LAST UPDATED: 10/5/2026
EMAIL: perry.beamer@gmail.com
pbeamer@ncsu.edu

EDUCATION

08/2023-Present	PhD. Biomathematics North Carolina State University (NCSU) GPA: 4.0
08/2018-05/2022	B.S. Mathematics <i>cum laude</i> & B.A. Philosophy <i>cum laude</i> University of Maryland, College Park (UMD) GPA: 3.927

RESEARCH INTERESTS

Mathematics/Statistics

Topological data analysis, topological machine learning, optimal transport, mathematical modeling

Biology

'Omics data analysis, molecular structure prediction, drug design.

PUBLICATIONS

Beamer, P., Cang, Z. *PHD-MS: Multiscale Domain Identification for Spatial Transcriptomics via Persistent Homology*. Cell Reports Methods. May 18, 2026. In press. doi: 10.1016/j.crmeth.2026.101376. Epub 2026 Mar 30.

Beamer, P., T. Byer, J. Wang, and Z. Cang. *Protein-ligand binding-affinity prediction via spatially-aware topological graph*. Paper in progress.

AWARDS AND HONORS

Lucas Service Award May 2026
Awarded for outstanding service to the NCSU Biomathematics community.

NCSU Biomath First-year Award May 2024
Awarded to most outstanding first-year Biomathematics PhD student at NCSU for academic performance

NCSU Graduate Fellowship August 2023
Awarded to best incoming graduate students in each department at NCSU

PRESENTATIONS

Oral

- Lightning talk, Triangle Computational and Applied Mathematics Symposium, NCSU November 2025
- **Invited Talk** Triangle Area Graduate Mathematics Conference, Duke University February 2025
- **Finalist**, DRUMS REU 3-minute thesis competition, NCSU July 2021
- Speaker, monthly FDA Division of Pharmacometrics meeting July 2020

Posters

- Society for Mathematical Biology, Graz, Austria. July 2026
- Triangle Computational and Applied Mathematics Symposium November 2025
- Society for Mathematical Biology, Edmonton OR. July 2025
- Integrative Symposium of NC State Training Grants, NCSU January 2025
- Joint Mathematics Meetings, virtual. April 2022
- Undergraduate Research & Creativity Symposiums, NCSU, Raleigh NC July 2021

TEACHING EXPERIENCE

Recitation Leader, MA141 NCSU Fall 2026
Taught 3 Calculus I recitation sections, totaling more than 80 students. Included grading exams.

Lecture Assistant, MA111 NCSU Fall 2025

Assisted with lecture and administered exams for 30 students in MA111 (precalculus). Included 5 weekly tutoring hours in the math tutoring center, including students in all NCSU math courses.

Math instructor, the Family Place, Washington D.C. August 2022-June 2023
Taught 11 classes of English as a second language (ESL) adults, including more than 100 students. Prepared students for Comprehensive Adult Student Assessment Systems (CASAS) exams, including topics from elementary-level arithmetic up to high-school-level algebra.

Private math tutor 2019-present
Tutored high-school and college students in Algebra I-II, Geometry, Precalculus, and Calculus I-III.

SERVICE

President, Biomathematics Graduate Student Association Fall 2025-Present
Led graduate student association for graduate program of 30 students. Spearheaded the introduction of a mentorship program for first-year graduate students. Other duties included organizing meetings and social events, managing budget, and planning outreach/recruitment efforts.

Graduate Assistant, DRUMS REU at NCSU Summer 2024
Supervised team of undergraduate researchers under the direction of Dr. Zixuan Cang. I managed tasks and daily workload for each student, helping develop a new method to analyze downstream genes affected by cell-cell communication.

Organizer and Judge, Triangle competition in mathematics modeling 2023-24
Recruited the NCSU team for undergraduate modeling competition between NCSU, Duke and UNC Chapel Hill. Increased NCSU participation significantly over previous years. Judged completed submissions for mathematical merit and clarity.

Club President, UMD Philosophy Club 2020-2022
Planned weekly discussions and social events for more than 150 club members. Advertised club activities on social media and organized virtual meetings during COVID-19 pandemic.

UNDERGRADUATE RESEARCH EXPERIENCE

05/2021-2022	Directed Research for Undergraduates in Mathematics & Statistics (DRUMS) North Carolina State University (NCSU) Department of Mathematics Mentors: Dr. Mette Olufsen & Dr. E. Benjamin Randall Title: <i>Using modeling and machine learning to characterize POTS</i>
05/2020-08/2020	Oak Ridge Institute for Science and Education (ORISE) Fellowship US Food and Drug Administration (FDA) Mentors: Dr. Yaning Wang & Dr. Ye Yuan Title: <i>Modeling daily state-level COVID-19 cases with a compartmental system of ordinary differential equations.</i>

SCIENTIFIC COURSEWORK

Mathematics/Statistics

Machine Learning, Mathematical Modeling, Algebraic Topology, Matrix Theory, Bioinformatics.

SKILLS

Computing

Proficient in MATLAB, R, Python, Java, $\text{\LaTeX}$, Microsoft Office.
Experience with JavaScript, HTML, FORTRAN.

Languages

Native speaker in English. Proficient in Spanish (10yrs experience) and Chinese (15yrs)